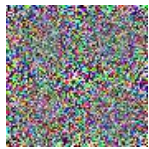


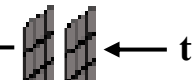
Hutchinson Trace Estimator (GCE ❄️)

Input $x_T \sim \mathcal{N}(0, I)$



$$\mathbb{E}_{t, x_0, \epsilon} \lambda(t) \|\epsilon - \epsilon_\theta(x_t(\phi), t)\|^2$$

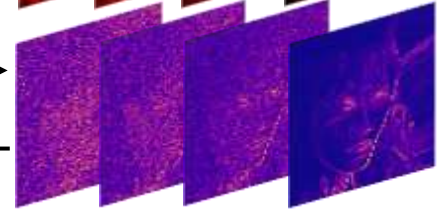
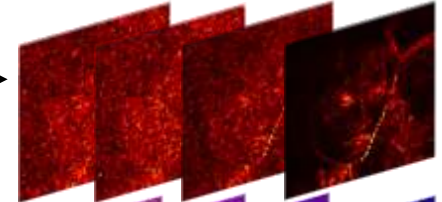
$$L_{\text{total}} = L_{\text{score}}(\theta, \phi) + L_{\text{GCE}}(\psi)$$



t

$$\gamma_t^2 = \text{Clamp}[\gamma_{\min}, \gamma_{\max}] \quad \psi_0 + \psi_1 \log(1 + |\hat{C}_t|)$$

$$\text{tr}(\nabla_x^2 \log p_t(x)) = \frac{1}{M} \sum_{i=1}^M \nu_i^\top (\nabla_x s_\theta) \nu_i$$



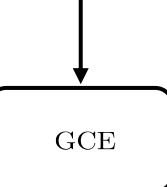
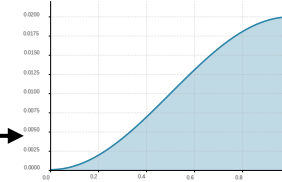
(GT)

Geometric Complexity Heatmap

t

$$e_t = \text{PE}(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \sin(\omega_2 t), \cos(\omega_2 t), \dots] \beta_t$$

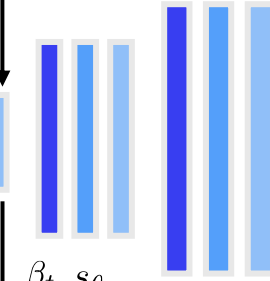
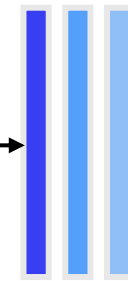
LNS



$$s_\theta(x_t, t) = \nabla_x \log p_t(x_t)$$



FiLM(γ^2)



β_t, s_θ



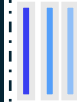
Training only



MLP



Element-wise summation



U-net Attention

$$x_t \in \mathbb{R}^{C \times H \times W} \quad t \in \{T, \dots, 1\} \quad e_t$$

$$C_t = \text{GCE}_\psi(x_t, t)$$

$$\gamma_t^2 = \text{Clamp}[\gamma_{\min}, \gamma_{\max}] \quad \psi_0 + \psi_1 \log(1 + |\hat{C}_t|)$$



Preconditioned Sampler

$$g_t = \gamma_t^2 \odot s_\theta(x_t, t)$$

$$x_{t-1} = x_t + \frac{1}{2} \beta_t \Delta t \cdot g_t - \frac{1}{2} \beta_t \Delta t \cdot (x_t + \nabla_x \gamma_t^2 + \gamma_t \odot \beta_t \Delta t \odot z), \quad z \sim \mathcal{N}(0, I)$$

$$\Delta t = \min(\Delta t_{\max}, \frac{C}{\max \gamma_t}) \quad \text{Repeat } t \rightarrow t-1, \quad \text{until } t=0, \quad \text{output } x_0$$

